

Elastic Stack (ELK) Project

Phase 1 & Phase 2 – Complete Command Guide (Without Git)

Update system

```
sudo apt update && sudo apt upgrade -y
```

This command updates the package list and upgrades all installed packages to their latest versions. It ensures system stability, security patches, and compatibility before installing Elastic Stack components.

Install Java

```
sudo apt install openjdk-17-jdk -y
```

Elasticsearch requires Java to run. This command installs OpenJDK 17, which is the recommended and supported Java version for Elasticsearch 8.x.

Verify Java installation

```
java -version
```

Verifies that Java is installed correctly and confirms the active Java version used by the system.

Install Elasticsearch

```
sudo apt install elasticsearch -y
```

Installs Elasticsearch, which acts as the core engine for storing, indexing, and searching logs collected from the system.

Start Elasticsearch service

```
sudo systemctl start elasticsearch
```

Starts the Elasticsearch service so that it begins accepting requests and indexing data.

Enable Elasticsearch on boot

```
sudo systemctl enable elasticsearch
```

Ensures Elasticsearch automatically starts whenever the system reboots.

Check Elasticsearch status

```
sudo systemctl status elasticsearch
```

Displays the current status, uptime, and health of the Elasticsearch service.

Test Elasticsearch API

```
curl -k https://localhost:9200
```

Tests whether Elasticsearch is running and reachable over HTTPS by querying its REST API endpoint.

Install Kibana

```
sudo apt install kibana -y
```

Installs Kibana, which provides a web-based interface to visualize, search, and analyze logs stored in Elasticsearch.

Start Kibana service

```
sudo systemctl start kibana
```

Starts the Kibana service so the web interface becomes accessible.

Enable Kibana on boot

```
sudo systemctl enable kibana
```

Ensures Kibana starts automatically after every system reboot.

Check Kibana status

```
sudo systemctl status kibana
```

Verifies whether Kibana is running properly or if any startup issues exist.

Access Kibana UI

```
http://localhost:5601
```

Opens the Kibana web dashboard in a browser for log exploration and visualization.

Install Filebeat

```
sudo apt install filebeat -y
```

Installs Filebeat, a lightweight log shipper used to collect and forward system logs to Elasticsearch.

Edit Filebeat configuration

```
sudo nano /etc/filebeat/filebeat.yml
```

Opens the Filebeat configuration file where log sources (inputs) and destinations (outputs) are defined.

Configure SSH authentication logs

```
/var/log/auth.log
```

Specifies the SSH authentication log file so Filebeat can monitor login attempts, failures, and possible brute-force activities.

Enable console output for testing

```
output.console: pretty: true
```

Configures Filebeat to print collected logs directly to the terminal. This is useful for testing before sending logs to Elasticsearch.

Test Filebeat configuration

```
sudo filebeat test config
```

Validates the Filebeat configuration file and checks for syntax or logical errors.

Run Filebeat in debug mode

```
sudo filebeat -e
```

Runs Filebeat in the foreground and displays live logs, helping verify that events are being collected correctly.

Start Filebeat service

```
sudo systemctl start filebeat
```

Starts Filebeat as a background service for continuous log collection.

Enable Filebeat on boot

```
sudo systemctl enable filebeat
```

Ensures Filebeat automatically starts after system reboot.

Check Filebeat status

```
sudo systemctl status filebeat
```

Checks whether Filebeat is running properly and actively harvesting logs.

Generate failed SSH login attempt

```
ssh fakeuser@localhost
```

Creates failed SSH login attempts, which are useful for testing log ingestion and detection scenarios.

View logs in Kibana

Kibana → Discover → filebeat-*

Allows analysis and visualization of collected SSH logs within Kibana's Discover section.